Software Requirements Specification

for

Student Activity Points Tracker (APP)

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
Group-6	12 Jan 2026	Initial Proposal	Ver 1.0
Group-6	20 Jan 2026	Grammatical Errors	Ver 1.1

1. Introduction

1.1 Purpose

The purpose of this document is to describe the requirements of the Student Activity Points Tracker Application. This system provides a digital solution for managing, verifying, and tracking student activity points in an educational institution. It eliminates manual paperwork and ensures transparency, accuracy, and efficiency.

1.2 Document Conventions

- *Section headings follow a **hierarchical numbering format** (e.g., 1, 1.1, 1.1.1) to clearly represent the structure and organization of the Activity Points Tracker App requirements.*

- *All **functional and non-functional requirements** are specified using standard requirement keywords to ensure clarity and consistency.*

- *The keywords **shall** and **may** are used in accordance with **IEEE standards**, where:*

    - ***Shall** indicates a **mandatory** system requirement*

    - ***May** indicates an **optional** feature or behaviour*

- *Important system components, modules, and features such as **User Authentication**, **Activity Submission**, **Faculty Review**, **Approval Workflow**, and **Dashboard** are highlighted in **bold** when first introduced.*

- *User roles including **Admin**, **Faculty Advisor**, **Student** are **capitalized** throughout the document to clearly distinguish them from general terms.*

1.3 Intended Audience and Reading Suggestions

- ***Students:***
  Students who use the Activity Points Tracker App to submit activity points proof, track approval status, view feedback, and monitor their overall activity points progress. Students should focus on the Overall Description, System Features, and Functional Requirements related to student activities.

- ***Faculty Advisors:***
  Faculty Advisors who review, verify, and approve student activities and provide feedback. Faculty Advisors should review the Overall Description and the Functional Requirements related to activity review, approval workflows, and student monitoring.

- ***Administrators:***
  System Administrators who manage user accounts, roles, permissions, system configuration, and

system maintenance. Administrators should review the System Overview, Administrative Functional Requirements, and Non-Functional Requirements sections.

- ***Developers:***
  Software developers responsible for designing, implementing, testing, and maintaining the Activity Points Tracker App. Developers should read the document sequentially, with particular emphasis on the Functional Requirements, Non-Functional Requirements, System Architecture, and System Interfaces sections.

1.4 Product Scope

- *The Activity Points Tracker App is an Android based application developed to automate and manage the submission, verification, and tracking of student activity points for all the students.*
- *Its primary purpose is to reduce manual effort, improve efficiency, and ensure a transparent and reliable activity evaluation process.*
- *The system benefits Students, Faculty Advisors, and Administrators by streamlining activity submission, review, approval, and management workflows.*
- *This software supports institutional academic goals by improving accountability, data accuracy, and overall academic activity management efficiency.*

1.5 References

- ***Software Requirements Specification (SRS)*** *Standard IEEE Std 29148-2018*
- ***Flutter Framework Documentation*** *Flutter SDK Documentation*
- ***Firebase Documentation*** *Firebase Authentication, Cloud Firestore, and Cloud Messaging Documentation*
- ***Activity points manual****, given by the institute.*

2. Overall Description

2.1 Product Perspective

- *This product is not part of any existing software suite and is being developed as a fresh solution to address the limitations of manual, paper-based, or informal methods used for tracking student activities and activity points.*
- *The application functions as a self-contained system that provides features for activity submission, verification, approval, point calculation, and record management.*
- *It directly interacts with external users such as Students, Faculty Advisors, and Administrators through dedicated user interfaces, relying on Firebase, etc.*

2.2 Product Functions

The application must facilitate the following key functions:

1. User Authentication and Access Control

Users must be able to securely access the Activity Points Tracker App based on their assigned roles.

- ***Signup and Login:*** *Users register and log in using authorized institutional credentials.*
- ***Role-Based Access Control:*** *Students, Faculty Advisors, and Administrators are granted different access permissions based on their roles.*
- ***Profile Management:*** *Users can view basic personal and academic information.*
- ***Password Recovery:*** *Secure mechanisms are provided to reset forgotten passwords.*

2. Activity Category and Policy Management

Administrators manage institutional activity definitions and rules.

- ***Activity Category Setup****: Admins define activity types, categories, and eligibility criteria.*
- ***Point Policy Configuration:*** *Activity point values and calculation rules are maintained.*
- ***Submission Window Control:*** *Administrators configure valid activity submission periods.*

3. Student Activity Submission

Students interact with the system to record their activities.

- ***Activity Entry:*** *Students submit activity details including title, category, date, and description.*
- ***Document Upload:*** *Supporting certificates and proof documents are uploaded digitally.*
- ***Submission Status View:*** *Students can track the status of each submitted activity.*

4. Activity Verification and Approval

Faculty Advisors evaluate submitted activities.

- ***Submission Review:*** *Faculty Advisors review activity details and attached documents.*
- ***Approval Decision:*** *Activity Points can be approved or rejected with remarks.*
- ***Faculty Monitoring:*** *Faculty Advisors view activity points progress of assigned students.*

5. Activity Point Calculation and Tracking

The system manages student activity scores.

- ***Automatic Point Calculation:*** *Points are computed based on approved activities and rules.*
- ***Cumulative Tracking:*** *Total activity points are updated and maintained per student.*
- ***Progress Overview:*** *Students and Faculty Advisors can view activity point summaries.*

6. Administration and Maintenance

Administrative functions ensure system reliability.

- ***User Management:*** *Administrators manage user accounts and role assignments.*
- ***Data Management:*** *The system maintains accurate records of activities and approvals.*

- ***System Maintenance:*** *Configuration and routine maintenance features are provided.*

2.3 User Classes and Characteristics

The Activity Points Tracker App is intended to be used by multiple user classes, each with distinct responsibilities, access privileges, and usage patterns. These user classes are differentiated based on frequency of use, functional access, technical expertise, and security levels. The major user classes are described below:

- ***Students:***
  Students are the primary users of the system. They use the application to submit activity details, upload supporting documents, and track the status and points of their activities. Students possess basic technical knowledge and have limited access privileges restricted to student-related functionalities only.

- ***Faculty Advisors:***
  Faculty Advisors use the system to review, verify, and approve or reject student activity submissions. They interact regularly with the system and have moderate technical expertise, with access limited to features related to assigned students and activity evaluation.

- ***Administrators:***
  Administrators are responsible for managing user accounts, roles, activity categories, system configurations, and maintenance activities. They have the highest level of access and technical expertise, but interact with the system less frequently compared to Students and Faculty Advisors.

2.4 Operating Environment

The Activity Points Tracker App is developed as an ***Android Application*** *designed to operate on commonly used smartphones and tablets. The system is intended to function reliably within standard mobile environments while coexisting with other applications on user devices.*

- ***Hardware Platform:***
  The application runs on standard mobile devices such as smartphones and tablets with adequate processing power, memory, and storage. No additional or specialized hardware is required.

- ***Operating System:*** *Compatible with Android 8.0 or above.*

2.5 Design and Implementation Constraints

- ***Platform Constraint***

  o The application shall be developed initially as an ***Android/iOS mobile application*** *and must support* ***Android version 8.0/iOS 12 and above****.*

- ***Technology Constraints***

  o The system shall be developed using ***Flutter*** *as the mobile application framework.*

  o Firebase *shall be used for backend services including authentication, database management, and*

cloud storage.

- ***Hardware Constraints***

  oThe application must operate efficiently on standard Android and iOS devices with limited memory, storage, and processing capabilities.

  *o Performance shall be optimized to function smoothly on **mid-range smartphones**.*

- ***Network Constraints***

  o A stable internet connection is required for user authentication, activity submission, verification, and data synchronization operations.

- ***Security Constraints***

  *o User authentication and data access shall be secured using **Firebase security rules** and **role-based access control** mechanisms.*

  o Sensitive student data and uploaded documents must be protected against unauthorized access and data breaches.

- ***Maintenance and Standards Constraint***

  *oThe application shall follow standard **Flutter coding conventions** and **clean architecture principles** to support future maintenance, scalability, and enhancements.*

2.6 User Documentation

The mentioned documentations will be provided later:

- *User Manual describing all features and workflows*
- *Quick start guide for Students and Faculty Advisors*
- *Administrator configuration and management guide*

2.7 Assumptions and Dependencies

Assumptions

- *It is assumed that all users (Students, Faculty Advisors, and Administrators) have access to compatible Android mobile devices and possess basic familiarity with using mobile applications.*

- *It is assumed that users will have a reliable internet connection to utilize system functionalities such as user authentication, activity submission, verification, and data synchronization.*

- *The system assumes the continued availability, maintenance, and support of Flutter and Firebase technologies throughout the development, deployment, and operational phases.*

- *It is assumed that institutions adopting the application will follow a standardized activity evaluation and approval process, allowing the system to be deployed with minimal customization*.

Dependencies

- *The application depends on Firebase services (Authentication, Database, Cloud Storage) for backend functionality and data management.*
- *The system depends on the Android/iOS operating system (Android (8.0) or iOS 12 and above) for successful deployment and execution.*
- *Proper functioning of the application depends on Google Play Services and Firebase infrastructure availability.*
- *Any changes in Firebase policies, pricing, or service availability may impact system functionality.*

3. External Interface Requirements

3.1 User Interfaces

*The Activity Points Tracker App provides a mobile-based graphical user interface (GUI) designed to be intuitive, consistent, and role-specific. The user interface facilitates interaction between the software system and different classes of users, namely **Students**, **Faculty Advisors**, and **Administrators**.*

Role-Based User Interfaces

- ***Student Interface:***
  Provides screens for activity submission, document upload, viewing submission status, tracking accumulated activity points, and managing personal profile information, with access restricted to the student's own records.

- ***Faculty Advisor Interface:***
  Provides screens for reviewing student activity submissions, approving or rejecting activities with remarks, and monitoring the activity progress of assigned students only.

- ***Administrator Interface:***
  Provides screens for managing user accounts, assigning roles, configuring activity categories and policies, and performing system configuration and maintenance tasks with full access privileges.

3.2 Hardware Interfaces

The Activity Points Tracker App interacts with standard mobile device hardware components to support user interaction, data capture, and communication. The hardware interfaces are described below:

- ***Supported Device Types:***
  The application supports Android and iOS based smartphones and tablets running Android version 8.0 or iOS 12 and above.

- ***Input Devices:***
  User input is captured through standard mobile hardware components such as touchscreens, virtual keyboards, and built-in navigation controls.

- ***Output Devices:***
  The application presents information through the mobile device display and provides user feedback using visual cues, audio alerts, and vibration features supported by the device.

- ***Network Interface:***
  The system utilizes the device's Wi-Fi or mobile data capabilities to communicate with cloud-based backend services. Internet connectivity is required for user authentication, data synchronization, and system updates.

- ***Notification Interface:***
  The application interfaces with the device's notification system, along with Firebase Cloud Messaging

(FCM), to deliver real-time alerts related to activity submission status, approvals, and system notifications.

- ***Data Interaction:***
  All data exchange between the mobile application and backend services occurs through secure network interfaces,without requiring direct or persistent interaction with device-specific hardware components.

3.3 Software Interfaces

Component	***Description***
Android or iOS	*Provides the runtime environment for executing the Activity Points Tracker mobile application and accessing device-level services.*
Flutter SDK	*Used to develop the mobile application user interface and handle client-side logic in a cross-platform manner.*
Firebase Authentication	*Manages secure user authentication for Students, Faculty Advisors, and Administrators using email and password credentials.*
Firebase Cloud Firestore	*Acts as the primary cloud database for storing user profiles, activity submissions, activity categories, approval status, and accumulated activity points in real time.*
Firebase Cloud Messaging (FCM)	*Used to deliver push notifications such as activity approval status updates, rejections, reminders, and system alerts.*
Firebase Cloud Storage	*Stores uploaded supporting documents and certificates submitted by students for activity point verification.*
Google Play Services	*Provides essential background services required for Firebase integration and reliable application functionality on Android devices.*
RESTful APIs (Firebase SDK)	*Enables secure communication between the Flutter frontend and Firebase backend services for data access and synchronization.*
JSON Data Format	*Used as the standard data format for all requests and responses exchanged between the mobile application and backend services.*

3.4 Communications Interfaces

The Activity Points Tracker system relies on the following communication protocols and interfaces to ensure secure and reliable data exchange between the mobile application and backend services:

- ***HTTPS (SSL/TLS Encryption):***
  All data transmitted between the mobile application and backend services shall be secured using HTTPS with SSL/TLS encryption to protect sensitive user information and prevent unauthorized access.

- ***RESTful APIs:***
  The application shall utilize RESTful APIs provided by Firebase SDKs to enable communication between the Flutter-based frontend and backend services.

- ***JSON Data Format:***
  All request and response messages exchanged between the frontend and backend services shall use the JSON (JavaScript Object Notation) format to ensure lightweight and structured data transfer.

- ***Firebase Cloud Messaging (FCM):***
  The system shall use Firebase Cloud Messaging to deliver real-time notifications such as activity submission status updates, approvals, rejections, reminders, and system alerts.

- ***Internet Connectivity:***
  A stable internet connection is required for complete application functionality, including user authentication, data synchronization, and real-time notification delivery.

4. System Features

4.1 User Authentication and Authorization

4.1.1 Description and Priority:

This feature allows users to securely access the system using role-based authentication. It ensures that only authorized users can access the system and are provided interfaces based on their role (Student, FA, Admin).

***Priority**: High.*

4.1.2 Stimulus/Response Sequences

- *User enters login credentials → System validates credentials and grants access.*
- *User requests password change/reset → System verifies identity and updates credentials.*

- *Admin assigns or updates user roles → System applies corresponding permissions.*

4.1.3 Functional Requirements

- ***FR-1.1:*** *Users shall log in and log out using valid credentials.*
- ***FR-1.2:*** *Users shall be redirected to their respective dashboards based on their role (Student, FA, Admin) after successful login.*
- ***FR-1.3:*** *The system shall provide role-based access for Students, Faculty Advisers, and Admins.*
- ***FR-1.4:*** *Users shall view their profile details.*
- ***FR-1.5****: If the credentials are invalid, the system shall display an error message.*
- ***FR-1.6****: Users shall change the default password during first-time login.*
- ***FR-1.7:*** *Users shall reset their password using registered email or mobile number.*

4.2 Student Certificate Upload & Verification

4.2.1 Description and Priority

Enables students to upload certificates related to academic, co-curricular, or extracurricular activities for record maintenance, verification, and progress tracking within the Activity Tracker Application.

Priority: *High (Essential for activity validation and student profile completeness)*

4.2.2 Stimulus / Response Sequences

- *Student logs in → System displays profile and uploaded certificates.*
- *Student uploads a certificate → System validates file format and size, then stores the document.*
- *Student updates or deletes a certificate unless it is approved → System updates records accordingly.*
- *FA reviews uploaded certificate → System marks certificate as verified or rejected*

4.2.3 Functional Requirements

- ***FR-2.1:*** *Students shall upload activity-related certificates.*
- ***FR-2.2:*** *Students shall upload certificates in common document formats such as PDF, JPG, and PNG.*
- ***FR-2.3:*** *The system shall validate mandatory fields, file size, and file type before accepting uploads.*
- ***FR-2.4:*** *Students shall update or delete uploaded certificates before verification by the Faculty Adviser.*
- ***FR-2.5:*** *Students shall view their total activity points and goal completion status.*
- ***FR-2.6:*** *Students shall view the certificate status as Pending, Verified, or Rejected.*
- ***FR-2.7:*** *The system shall prevent duplicate submissions for the same activity by a student*

4.3 Faculty Advisor (FA) Management Features

4.3.1 Description and Priority

This feature allows Faculty Advisors to manage and verify student activity submissions. It supports evaluation of applications, proof validation, and updating of student activity points.

Priority: *High.*

4.3.2 Stimulus/Response Sequences

- *FA opens student list → System displays assigned students and total activity points.*
- *FA opens application → System displays proofs and details for review.*
- *FA approves/rejects → System updates points accordingly and records action.*

4.3.3 Functional Requirements

- ***FR-3.1:*** *Faculty Adviser (FA) shall view assigned students and their total activity points.*
- ***FR-3.2:*** *FA shall access detailed activity submissions and view or download uploaded proof files.*
- ***FR-3.3:*** *FA shall approve or reject activity submissions and shall provide remarks when rejecting a submission.*
- ***FR-3.4:*** *The system shall automatically update student activity points after approval by the Faculty Adviser.*

4.4 Activity Category & Type Management

4.4.1 Description and Priority

Manages different activity categories and activity types to organize user activities efficiently.

Priority: *High (Core structure for activity tracking)*

4.4.2 Stimulus / Response Sequences

- *Admin creates activity categories → System stores and displays them.*
- *User selects an activity category → System loads related activity types.*

4.4.3 Functional Requirements

- ***FR-4.1:*** *Users shall view multiple activity categories and the respective points for each category.*
- ***FR-4.2:*** *Students shall select an activity category while submitting an activity.*
- ***FR-4.3:*** *Admins shall add, edit, or remove activity types.*
- ***FR-4.4:*** *The system shall display a confirmation message before deleting any activity category.*
- ***FR-4.5:*** *Admins shall define point criteria for each activity category*

4.5 Goal Setting & Progress Monitoring

4.5.1 Description and Priority

Allows Faculty Advisor to set activity point goals and monitor progress over time.

Priority: *High (Motivation and productivity tracking)*

4.5.2 Stimulus / Response Sequences

- *FA sets a goal → System saves goal details.*
- *Activity logged → System updates progress automatically.*

4.5.3 Functional Requirements

- ***FR-5.1:*** *FA shall define semester wise or yearly goals for students.*
- ***FR-5.2:*** *The system shall track progress against defined goals.*
- ***FR-5.3:*** *Users shall view goal completion status.*

4.6 Activity History & Analytics

4.6.1 Description and Priority

Provides users with historical views and basic analytics of their activities.

Priority: *Medium*

4.6.2 Stimulus / Response Sequences

- *User selects a date range → System displays activity history and statistics.*

4.6.3 Functional Requirements

- ***FR-6.1:*** *Users shall view activity application history sorted by date.*
- ***FR-6.2:*** *The system shall generate basic activity statistics based on submitted and verified activities.*
- ***FR-6.3:*** *Students and Faculty Advisers shall view visual summaries of activity data.*

4.7 Communication Notifications & Reminders

4.7.1 Description and Priority

Notifies users about missed activities, goals, or reminders through in-app notifications and enables communication between FA and students.

Priority: *Medium*

4.7.2 Stimulus / Response Sequences

- *Student submits activity application → System sends FA a notification.*
- *FA approves/rejects application → System sends Student a notification.*
- *Scheduled time reached → System sends reminder notification.*

4.7.3 Functional Requirements

- ***FR-7.1:*** *The system shall send reminders to students for pending activity submissions.*
- ***FR-7.2:*** *The system shall notify users about upcoming goal deadlines.*

- ***FR-7.3:*** *The system shall notify students when the Faculty Adviser approves or rejects an activity submission.*
- ***FR-7.4:*** *The system shall notify the Faculty Adviser when a student submits an activity application.*
- ***FR-7.5:*** *Users shall view past notifications in their notification history.*
- ***FR-7.6:*** *Students shall send queries to the Faculty Adviser through the system.*
- ***FR-7.7:*** *Faculty Advisers shall respond to student queries through the system*

4.8 Reports & Data Export

4.8.1 Description and Priority

Allows users to generate and export activity reports for review.

Priority: *Medium*

4.8.2 Stimulus / Response Sequences

- *User requests report → System generates and displays report.*

4.8.1 Functional Requirements

- ***FR-8.1:*** *The system shall generate semester-wise and year-wise activity reports.*
- ***FR-8.2:*** *Users shall download generated reports in standard file formats such as PDF and CSV.*

4.9 Administration & User Management

4.9.1 Description and Priority

Provides administrators with control over users, activity categories, and system data.

Priority: *Medium*

4.9.2 Stimulus / Response Sequences

- *Admin adds/removes users → System updates records.*

4.9.3 Functional Requirements

- ***FR-9.1:*** *Admin shall create, add, and modify user accounts.*
- ***FR-9.2:*** *Admin shall manage activity categories.*
- ***FR-9.3:*** *Admin shall assign or modify user roles.*
- ***FR-9.4:*** *The system shall restrict Admin from assigning or modifying student activity points.*

5. Nonfunctional Requirements

5.1 Performance Requirements

- *The application should load the **dashboard/home screen** within 3-5 **seconds** under a stable **4G or Wi-Fi** internet connection.*

  ***Rationale:** To provide a fast and responsive user experience by minimizing initial waiting time for users.*

- *Activity-related information such as **activity categories**, **submitted activities**, and **activity point summaries** should be displayed within **3-5 seconds** after a user request.*

  ***Rationale:** To enable users to quickly access activity details and summaries, ensuring smooth navigation and efficient interaction with the application.*

- *The system should support **simultaneous access by up to 100-300 users** without noticeable performance degradation.*

  ***Rationale:** To ensure system stability and consistent performance during peak usage periods with high user concurrency.*

- *Notifications related to **activity submission updates**, **approvals**, **rejections**, and **system alerts** should be delivered within **3–5 seconds** of the triggering event.*

  ***Rationale:** To ensure timely communication of activity status updates and alerts, keeping users informed in near real time.*

- *Critical operations such as **activity submission**, **faculty approval or rejection**, **activity point calculation**, and **status updates** should be completed within **5–7 seconds** under normal operating conditions.*

  ***Rationale:** To ensure essential system operations are processed efficiently without causing delays or user frustration.*

5.2 Safety Requirements

- ***Data Loss Prevention**:*
  The system shall ensure that student activity data, approval records, and activity points are not lost due to application crashes, network failures, or unexpected shutdowns. Automatic data synchronization and cloud storage mechanisms shall be used to minimize data loss.

- ***Unauthorized Access Prevention**:*
  The application shall prevent unauthorized access to sensitive user data by enforcing secure authentication, role-based access control, and session management mechanisms.

- ***Data Integrity Protection**:*
  The system shall prevent unauthorized modification or deletion of activity records, approval decisions, and activity point calculations. Only authorized users shall be permitted to perform such actions.

- ***Safe Handling of Uploaded Documents:***
  Uploaded certificates and supporting documents shall be securely stored and protected against accidental deletion, corruption, or unauthorized viewing.

- ***Error and Failure Handling**:*
  The system shall handle runtime errors and failures gracefully by displaying appropriate error messages and preventing incomplete or inconsistent data states.

5.3 Security Requirements

- ***Role-Based Access Control:***
  The application shall enforce role-based access control to ensure that users can only access features and data permitted by their assigned roles.

- ***Secure Communication:***
  All data exchanged between the mobile application and backend services shall be transmitted using secure communication protocols (HTTPS with SSL/TLS encryption) to prevent data interception or tampering.

- ***Data Privacy and Confidentiality:***
  The system shall protect sensitive user data, including personal details, activity records, and uploaded documents, from unauthorized access or disclosure.

- ***Secure Data Storage:***
  User data and documents shall be stored securely with controlled access and shall not be accessible to unauthorized users.

- ***Protection Against Unauthorized Actions:***
  The application shall prevent unauthorized actions such as activity approval, rejection, or modification by users without sufficient privileges.

- ***Compliance with Institutional Policies:***
  The system shall comply with institutional data protection and privacy policies governing student records and academic data.

5.4 Software Quality Attributes

- ***Availability:***
  The application shall ensure high system availability (around 80% uptime), except during planned maintenance activities.
- ***Reliability**:*
  The system shall handle failures gracefully and ensure that user actions are not lost during crashes or unexpected interruptions.
- ***Maintainability:***
  The application shall be designed in a modular manner to simplify maintenance, updates, and defect resolution.
- ***Usability:***
  The system shall provide an intuitive interface that allows users to perform common tasks efficiently with minimal effort.
- ***Scalability:***
  The application shall be capable of supporting an increasing number of users, institutions, and departments through a scalable cloud-based architecture.
- ***Portability:***
  The system shall operate on Android devices and be structured to allow future extension to other platforms with minimal modifications.
- ***Testability:***
  The application shall support automated testing mechanisms to verify correctness, stability, and overall system quality.

5.5 Business Rules

- *Only verified and authenticated users shall be allowed to access the application based on their assigned roles.*
- *Faculty Advisors may review and approve activities only for students officially assigned to them.*
- *Students shall be permitted to submit activity details only within the defined submission deadlines set by the institution.*
- *Activity status and point totals shall be updated automatically after an activity is approved or rejected.*
- *Approved activity submissions shall not be editable by students to preserve data integrity.*
- *Administrators shall be authorized to add, remove, or modify user accounts, roles, and activity configurations as required.*
- *All activity-related actions, including submissions, approvals, rejections, and modifications, shall be logged for monitoring, auditing, and accountability purposes.*

6. Other Requirements

6.1. Database Requirements

- *The system shall use a centralized cloud-based database to store user profiles, activity submissions, approval records, and activity point details.*
- *The database shall support concurrent access by multiple users without data conflicts.*
- *Secure access controls shall be enforced to protect stored data.*

6.2. Data Consistency Requirements

- *The system shall ensure consistency between activity submissions, approval status, and calculated activity points across all user views.*
- *Updates to activity status or points shall be reflected immediately across the system.*
- *The system shall prevent duplicate or conflicting records during concurrent operations.*

6.3. Internationalization Requirements

- *The application shall support English as the default language.*
- *The system shall be designed to allow future addition of other languages without major architectural changes.*

6.4. Legal and Compliance Requirements

- *The system shall comply with institutional policies related to student data privacy and academic records.*
- *Personal and academic data shall be collected, processed, and stored only for authorized institutional purposes.*
- *The application shall adhere to applicable data protection and privacy regulations relevant to educational systems.*

6.5. Reuse and Extensibility Requirements

- *The system architecture shall promote reuse of components across similar academic or activity management applications.*
- *The application shall be designed to support future enhancements, such as new activity categories or evaluation rules, with minimal code changes.*

Appendix A: Glossary

- ***Administrator (Admin):***
  Manages users, roles, system configurations, and ensures smooth operation of the application.
- ***Faculty Advisor (FA):***
  Reviews, monitors, and approves activities submitted by assigned students.
- ***Student:***
  A registered user who submits activities and tracks their approval status within the system.
- ***FCM (Firebase Cloud Messaging):***
  A service used to send real-time push notifications to users.
- ***Flutter:***
  A cross-platform framework used to develop the mobile application using a single codebase.

- ***Firebase:***
  A backend platform providing authentication, database services, and cloud messaging.

- ***SRS (Software Requirements Specification):***
  A formal document that defines the functional and non-functional requirements of the system.

Appendix B: Analysis Models

<Optionally, include any pertinent analysis models, such as data flow diagrams, class diagrams, state-transition diagrams, or entity-relationship diagrams.>

Appendix C: To Be Determined List

The following items are identified as To Be Determined (TBD) within this Software Requirements Specification. These items require further clarification, institutional decision-making, or implementation-time configuration and will be tracked to closure during later phases of system design and deployment.

TBD-1: *User Role Identification Mechanism*
The exact mechanism for identifying and assigning user roles (Student, Faculty Advisor, Administrator) during registration—such as email domain validation or admin approval—has not yet been finalized.

TBD-2: *Activity Categories and Limits*
The predefined categories of activities and the maximum number of activities a student may submit per academic period require confirmation from institutional authorities.

TBD-3: *Activity Submission Deadlines*
Exact timelines for activity submission and approval will be configured per academic term.

TBD-4: *Data Retention and Archival Policy*
The duration for which student activity records, approval history, and audit logs are retained before archival or deletion is subject to institutional policy.

TBD-5: *Mobile Platform Feature Scope*
The extent of functionality supported on mobile devices developed using Flutter (full access versus limited features) is yet to be finalized.

TBD-6: *Integration with Institutional Systems*
Future integration with existing academic or student information systems is currently undefined and will be addressed in later phases.